

BROKILD · WINDOWS VST3 · BINAURAL ROOM

THIN WALLS

HANDBOOK

BUILD 260925.5

Three rooms, three doors, four sources, whatever furniture you put in them, and a head you can walk around with. Everything you hear is a path the engine has actually traced — through the air, off the walls, over the sofa, round the door frame or straight through the leaf. [September 2026](#).



AN APARTMENT YOU CAN PUT A SOUND INSIDE

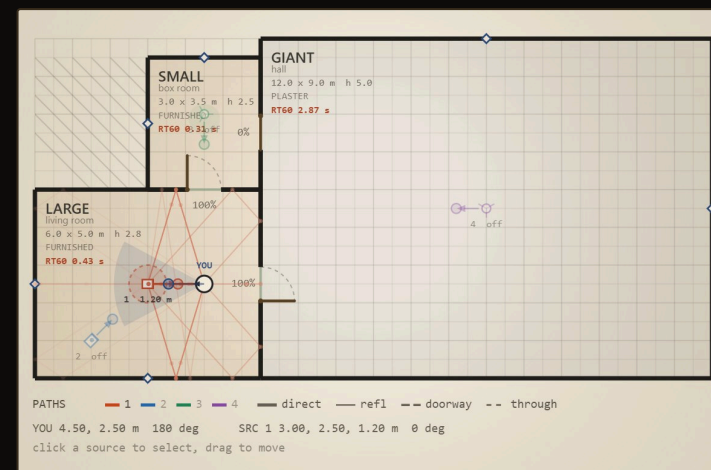
Thin Walls is a small flat: a **living room**, a **box room** and a **hall**, joined by three doors that you can open, shut or leave ajar. You place up to four sound sources in it, you stand in it yourself, and what comes out of the plugin is what your two ears would pick up from where you are standing and which way you are facing.

It is not a reverb with a room preset. Every arrival is computed as a journey: the straight line to each ear, the first and second bounces off the walls, the path that bends round a door frame, the muffled fraction that gets through a shut door or a party wall, and the tail each room builds for itself. Move the source a metre and the whole set of journeys changes, which is why close, far and next-door sound different in character and not merely in level.

There is no reverb-amount control, because the amount is decided by where you stood. You can furnish the rooms, hang your own pictures on the walls, and record a walk through the flat as a video with its own sound.

INSTALLING

Copy the whole **Thin Walls.vst3** folder into your VST3 folder — normally C:\PROGRAM FILES\COMMON FILES\VST3 — and rescan in your DAW. The standalone in the same download needs no installing; double-click it.

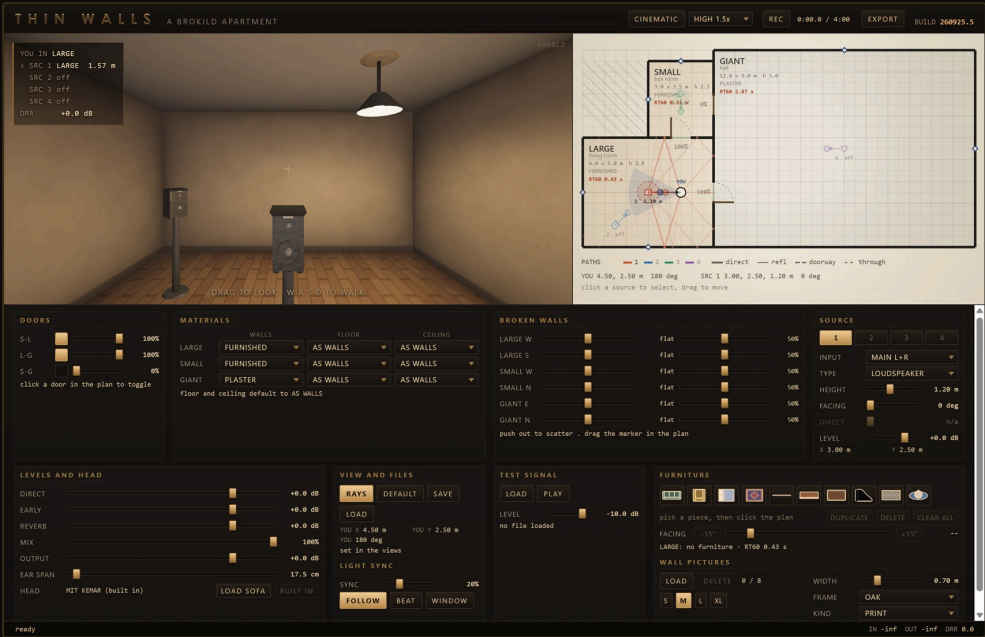


The flat. Living room bottom left, box room above it, hall to the east. The dark block top left is outside the apartment: nothing can stand there.

THE FIRST MINUTE

Put it on a track. Drag the loudspeaker glyph in the plan away from the head and back. Then click the door between the living room and the hall to shut it, and drag the source through into the hall. That is the whole instrument in three gestures.

TWO VIEWS AND A ROW OF CONTROLS



Top: CINEMATIC, REC and EXPORT, and the build. **Left:** the view from where you are standing. **Right:** the plan. **Below:** eight groups of controls, in two rows — the second row scrolls into view. **Foot:** input, output and the direct-to-reverberant ratio.

EVERY CONTROL EXPLAINS ITSELF

Hover any slider, button or menu and a panel appears naming it and saying what it is for. Nothing in this handbook is hidden from the instrument itself; the handbook is here for the things that need a picture or a table.

NOTHING IS MODAL

The two views and the controls are always live at once. A drag in the plan moves the 3D view as it happens, and walking in the 3D view moves the glyph in the plan. Both write the same host parameters, so both are recorded by your DAW's automation.

RAYS

The **Rays** button draws the engine's own paths on the plan: direct, first and second order reflections, doorway portals, and sound through a shut leaf. It is the fastest way to see why a move changed the sound.

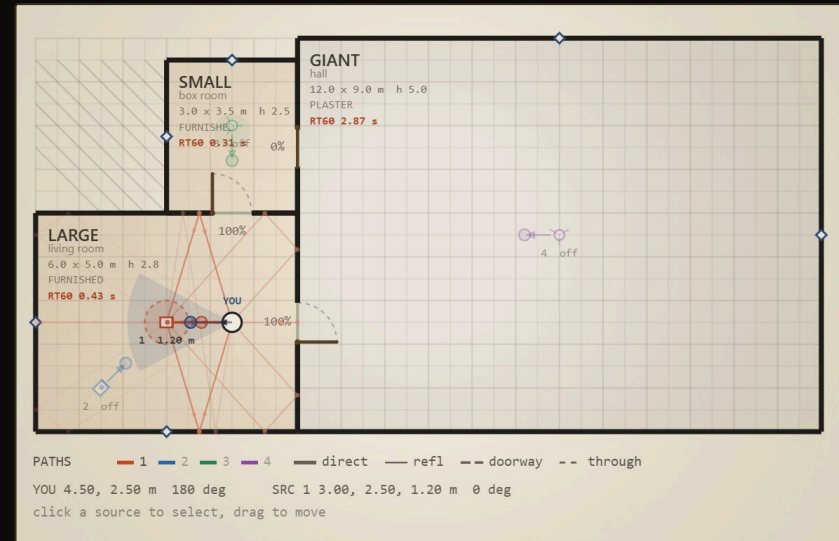
THE EIGHT GROUPS, AND THE HEADER

Doors	one aperture each, and a button for the two ends	Source	everything about whichever of the four is selected	Test signal	a file of your own, looped into the input
Materials	nine choices — walls, floor and ceiling of each room	Levels and head	the three trims, mix, output and ear spacing	Furniture	the tray of ten pieces, and the wall pictures
Broken walls	split a wall and push the break out, per room	View and files	rays, defaults, save and load, where you stand, and light sync	Header	CINEMATIC and its quality, REC, EXPORT

WALK, LOOK, DRAG



The 3D view is **you**. The camera is the listener's head, at 1.65 m. Drag inside it to look around; **W A S D** to walk. Walls stop you, so does a door less than half open, and so does any furniture that stands in the way of sound.



The plan is the **architect's drawing**. Drag any source or the head. Turn either by its heading handle. Click a source to select it for the controls below; click a door to toggle it, or drag along it to set it part open.

TURNING YOUR HEAD

Facing is not decoration. The binaural rendering is driven by the direction each arrival comes from **relative to your head**, so turning on the spot moves every source around you — including the ones you cannot see, behind a shut door.

WHERE YOU MAY STAND

Anywhere inside the three rooms. The area north-west of the living room is outside the flat and is solid; the plan will not let you drag there, and if automation sends you there the engine puts you back in the nearest room. Walking through an open doorway is seamless: for 0.4 m either side of the door the engine hears both rooms at once and hands one to the other (page 15).

TAKE HOLD OF THE HANDLE

Every door in the 3D view has a handle. Walk up to one — within **2.5 m**, in front of you and not through a wall — and it lights up as the pointer crosses it. **Click** to slam it or throw it wide; **drag** to swing it to anywhere in between. Grabbing a handle is not a look, so the room does not spin while you do it.

THE KEYS ALWAYS WIN

W A S D and the arrow keys walk and turn **even when a menu or a fader has the focus**. Set a material, then walk — the key never reaches the menu behind it. A key typed into a text field is still the field's.

THREE ROOMS, THREE DOORS

Fixed, and deliberately so: every room is a different size and every pair of rooms has a door, so every “behind that one” case exists, including the two-door route through the third room.

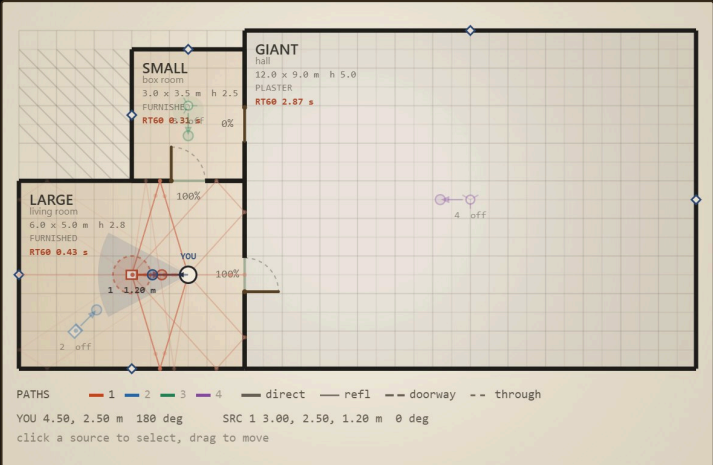
ROOMS

ROOM	SIZE (M)	HEIGHT	VOLUME WHAT IT IS
LARGE	6.0 × 5.0	2.8	84 m ³ the living room
SMALL	3.0 × 3.5	2.5	26 m ³ the box room
GIANT	12.0 × 9.0	5.0	540 m ³ the hall

DOORS

DOOR	BETWEEN	HINGED	DEFAULT
SMALL – LARGE	box room and living room	west side	open
LARGE – GIANT	living room and hall	south side	open
SMALL – GIANT	box room and hall	north side	shut

Every door is 0.90 m wide and 2.05 m high. Your ears are at 1.65 m; a source can stand between 0.2 and 2.4 m off the floor.



Origin is the south-west corner of the living room. X runs east, Y runs north. A facing of 0° is east and the angle increases anti-clockwise, so 90° is north.

THE TWO-DOOR ROUTE

Shut the living room door and the hall is not silent: it still reaches the box room through its own door, and from there back to you. Opening both doors of that route brings the sound up by **18 dB**. The path bends twice, and it is drawn.

FOUR OF THEM, AND TWO KINDS

Each source has its own position, height, facing, type, level and input. The numbered tabs pick which one the controls belong to; clicking a source in the plan picks it too.



LOUDSPEAKER — a modelled two-way monitor: a 6.5" woofer below 2.2 kHz, a 1" tweeter above. Nearly omnidirectional at 125 Hz, beaming with frequency in front, and about **22 dB down straight behind it at 4 kHz**.



PURE — a point source with no box, on a stand. DIRECTIVITY runs from omnidirectional to a cardioid and is the same at every frequency. The HEIGHT control is the stand.

FACING MATTERS

Turning a loudspeaker away from you takes **21.5 dB** off 4 kHz and only **5.5 dB** off 250 Hz. That is what a speaker pointed at the wall sounds like, and it is why aiming one into the hall and standing behind it is a usable effect rather than just a volume drop.

A DIRECTIONAL SOURCE DRIVES THE ROOM LESS

The room's tail is fed by the source's radiated **power**, not by what is aimed at you. Narrowing the pattern removes power, so a cardioid excites the room **4.8 dB** less than an omni at the same on-axis level. Point a speaker away and the room gets brighter relative to it, not darker.

HEIGHT

0.2 to 2.4 m. Low is floor-coupled and boomy, ear height is the most direct, high throws more into the ceiling. It also changes which ear is nearer, which is audible inside about a metre.

FEEDING THE FOUR

Every source has an **INPUT** menu with seven positions. Its first is **OFF**: the source makes no sound, costs nothing, and stays in the views, drawn solid but unlit — only a fed source glows — so you can place it before you use it. The other six take a channel from one of two stereo buses.

SETTING	WHAT FEEDS THE SOURCE
OFF	nothing — silent, unlit, still placeable
MAIN L / R	one channel of the track the plugin is on
MAIN L+R	both, summed
AUX L / R	one channel of the second input bus
AUX L+R	both, summed

The obvious use of MAIN L and MAIN R is a stereo pair: two loudspeakers, one fed each channel, standing where you would put them in a room. The obvious use of AUX is a second instrument in a different room, sharing one apartment.

ROUTING THE AUX BUS

The plugin declares a second stereo input bus named **Aux In**, disabled until the host enables it. Most DAWs call this a **sidechain**. In Ableton Live, open the plugin's device title bar, switch on **Sidechain** and pick the track you want to send; in Reaper, add channels 3–4 to the track and pin the other track's send into them; in Cubase and Studio One, use the plugin's own side-chain button and route a send to it.

The panel greys the AUX entries out until the host has actually enabled the bus, so a source can never be pointed at silence without saying so.



The **source** panel. The tabs across the top pick which of the four you are editing.

FOUR SOURCES, ONE APARTMENT

Sources in the same room share that room's tail, which is correct: one room has one reverberant field however many things are shouting into it. Sources in different rooms drive their own rooms and reach you through the doors.

One source costs about **18 %** of a core at 48 kHz. Four, with ten pieces of furniture in the flat, about **44 %**. Switching a source to OFF gives its share straight back.

WHAT THE WALLS ARE MADE OF

Five surface treatments, and any of them on any surface of any room. These are published octave-band absorption coefficients, not voicing presets, and the reverberation time follows from them.

THE FIVE

MATERIAL WHAT IT IS

ABSORBING	heavy drapes, thick carpet, soft furniture
FURNISHED	a lived-in living room: plaster, carpet, curtains
PLASTER	bare plaster on brick, wooden floor
TILED	glazed tiles, a little glass and wood
STUDIO	treated: broadband absorbers, bass traps and diffusers — page 10

Absorption rises with frequency on the first four, which is why a live room here gets darker as it decays rather than simply longer. ABSORBING takes 80 % at 4 kHz and only 20 % at 125 Hz, so a dead room still has a bloom in the bottom octave — as real ones do. STUDIO is the one that does not, and that is its whole point.

AN OPEN DOOR IS A HOLE

Sound that leaves through a doorway does not come back, so the engine counts an open door as a perfect absorber. The box room in bare plaster rings for **1.32 s** at 1 kHz with its doors shut and **0.58 s** with both wide open. You can hear a room change size by opening a door into it.

REVERBERATION TIME, DOORS SHUT

Seconds, from Eyring's formula with ISO 9613-1 air absorption — the same arithmetic the engine runs at every change. STUDIO is on [page 10](#).

ROOM	MATERIAL	125 HZ	500 HZ	1 KHZ	4 KHZ
LARGE	ABSORBING	0.51	0.13	0.09	0.07
LARGE	FURNISHED	1.07	0.65	0.51	0.34
LARGE	PLASTER	3.57	2.16	1.77	1.17
LARGE	TILED	5.24	5.18	3.42	2.07
SMALL	ABSORBING	0.38	0.10	0.07	0.06
SMALL	FURNISHED	0.78	0.48	0.38	0.25
SMALL	PLASTER	2.50	1.59	1.32	0.90
SMALL	TILED	3.59	3.83	2.57	1.64
GIANT	ABSORBING	0.92	0.23	0.15	0.13
GIANT	FURNISHED	1.94	1.16	0.90	0.58
GIANT	PLASTER	6.58	3.80	3.06	1.88
GIANT	TILED	9.78	8.90	5.75	3.06

Each door slider is an aperture, continuous from shut to wide open; the button beside it does the two ends. Clicking a door in the plan does the same thing, dragging along one sets it part open, and in the 3D view you can simply take hold of its handle.

WALLS, FLOOR AND CEILING, EACH ITS OWN

Nine choices, three for each room. Floor and ceiling start at AS WALLS, so a room you never touch is exactly the room it always was.

A real room is not made of one thing. Between a source standing on the floor and a listener standing on the floor, the **first two reflections to arrive are the one off the floor and the one off the ceiling** — they are the shortest detours there are. They arrive within a few milliseconds of the direct sound, which is exactly the window the ear uses to decide how big a room is and where in it you are.

So the floor and the ceiling are worth their own choices, and they are the two a real studio treats first.

A CARPET IS THE MOST LOPSIDED ABSORBER IN THE FLAT

Thick carpet takes almost nothing below 250 Hz and a great deal above 2 kHz. Setting **floor** to ABSORBING therefore kills the floor bounce and the flutter with it while leaving the bottom of the room exactly where it was — which is what laying a rug in a hard room really does, and why it is the first thing anyone tries.

A CLOUD OVER THE DESK

Ceiling set to ABSORBING is the cloud every control room has above the listening position. It is the opposite trade to a carpet: it takes the vertical flutter out and leaves the room's width and depth ringing.

MATERIALS

	WALLS	FLOOR	CEILING
LARGE	PLASTER	ABSORBING	ABSORBING
SMALL	FURNISHED	AS WALLS	AS WALLS
GIANT	STUDIO	AS WALLS	AS WALLS

floor and ceiling default to AS WALLS

Nine selectors. A row per room, a column per surface. AS WALLS is not a fifth material, it is a pointer: that surface follows whatever the walls are set to.

WHAT IT IS WORTH

The living room in bare plaster rings for **2.71 s** at 250 Hz and **1.77 s** at 1 kHz. Leave the walls in plaster and put a carpet down and a cloud up — two menu changes — and the same room measures:

SURFACES	250 HZ	1 KHZ	4 KHZ
plaster throughout	2.71	1.77	1.17
+ carpet and cloud	0.39	0.20	0.17

Read off the panel's own readout with the engine running. Two of the six surfaces carry most of the room.

STUDIO: TREATED, AND STILL LIVE

A room whose decay does not move with frequency, and whose treatment is diffusion rather than absorption.

The other four materials are measured surfaces. STUDIO is not: its absorption was **solved backwards from Eyring's formula** for the one property a control room is built to have — a reverberation time that is the same at the bottom of the band as at the top.

What comes out is **constant absorption near a quarter**, with a little relief at the very top to pay for the air absorption that takes the top octave anyway. That is not a coincidence: broadband absorbers plus bass traps are exactly what it takes to flatten a real room, and this is the coefficient curve they add up to.

ROOM AS A STUDIO	125 HZ	500 HZ	1 KHZ	4 KHZ	SPREAD
LARGE · living room	0.39	0.40	0.40	0.41	1.04×
SMALL · box room	0.29	0.29	0.30	0.31	1.06×
GIANT · hall	0.71	0.71	0.71	0.71	1.00×
GIANT · the same hall, TILED	9.78	8.90	5.75	3.06	3.09×

Spread is the ratio of the longest of 250 Hz, 1 kHz and 4 kHz to the shortest, measured on the rendered audio. The hall as a studio is **1.00×**: flat to the digit.

DIFFUSION, NOT ABSORPTION

The other four materials return a reflection in the specular direction and take a bite out of it. A studio's diffusers do something quite different: they **keep the energy and throw away the direction**.

So what comes back off a treated wall is still in the room — it just no longer arrives as a single sharp image, and it no longer bounces between two parallel walls in step with itself. Measured: the specular reflections carry **17.6 %** of what the same room in plaster returns, and the rest is scattered into the room rather than absorbed out of it.

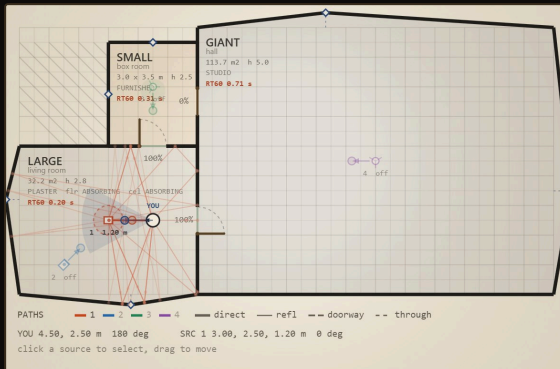
WHICH IS WHY IT IS NOT DEAD

A hall that rings for 0.71 s is a live room, not an anechoic chamber. You can still hear that it is big. What you cannot hear is a flutter echo, or the bottom end arriving late: the early response of a centred source measures **1.1 dB** uneven across the middle of the band, where the same room in plaster measures **2.4 dB**.

STUDIO also splays its walls very slightly, which is the same trick the next page gives you by hand — and the two add up.

THE ROOMS ARE NOT SHOEBOXES ANY MORE

Two walls of every room can be split at a movable point and that point pushed out, so the room stops being rectangular. Drag the diamond in the plan, or use the two faders.



Four walls broken. The diamond on each is the break: drag it along the wall, or out of it. Area, volume and reverberation time all follow — the hall's floor goes from 108.0 to **113.7 m²**, and its own decay from 1.766 s to **1.801 s**.

PUSH IT OUT

SPLAY

Two panels that are no longer parallel to the wall facing them cannot sustain a flutter between them, and they spread a reflection over a range of angles instead of returning it whole. Off centre is worth more than centred: two panels of different lengths scatter over a wider band than two of the same length.

DO NOT PULL IT IN

CONCAVE FOCUSES

Pushed the other way the wall bends **into** the room, and a concave surface gathers sound to a point instead of spreading it. That is worse than leaving the wall flat — real rooms are splayed outward, never inward. The plan says so when you do it. It is offered because it is audible and instructive, not because it is good.



Two faders per wall. How far the break is pushed (± 0.6 m, **flat** at the centre) and where along the wall it sits. The two walls offered are always the ones with no doorway, so a break can never collide with a door.

WHAT IT REALLY DOES

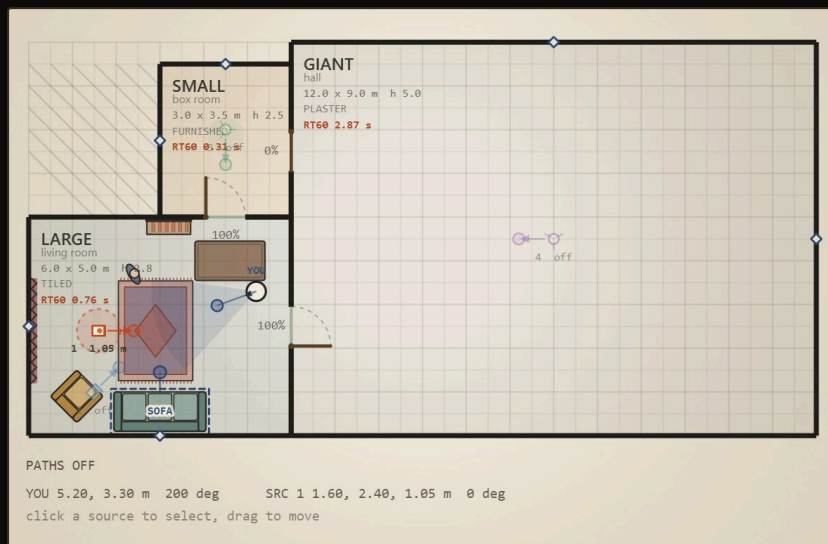
HONESTLY

In this model the splay acts on the **early reflections**, not on the room's standing waves: the late field is statistical rather than modal. Above the Schroeder frequency — about 72 Hz in the hall as a studio — that is where splay does its audible work in a real room too. Below it, absorption is what helps.

A room that is not a box needs a different search. A room is now a list of planar surfaces and a path a sequence of them, traced back from the listener and validated at every step; a square room still takes the exact arithmetic. A wall broken by **two millimetres** is the same room, and the general search reproduces the exact one to **0.09 dB** overall and **0.20 dB** over the first 50 ms.

FURNISH THE FLAT

Ten pieces, up to 24 in the apartment, placed and turned in the plan and built in the 3D view. Every one is heard as well as seen.



The plan. A rug, a sofa (selected, with its turn handle on its front), an armchair, a bookcase, a table, a curtain on the west wall and a standing person. The room's RT60 under its name is the engine's, furniture included.

THE TRAY

CLICK, THEN CLICK THE PLAN

The ten pieces sit as a row of icons at the top of the **Furniture** group: sofa, armchair, bed, rug, curtain, bookcase, table, grand piano, wardrobe and a person. Click one to arm it and click in the plan to set it down — or drag it straight onto the plan.

PICK UP, TURN, COPY

IN THE PLAN

Click a piece to select it and drag it about. Turn it by the round handle on its front, in 15° steps (Shift lets go of them), or with **FACING** and its -15° / +15° buttons. **DUPLICATE** sets a copy beside it; **DELETE** or the Delete key takes it out; Escape lets go. **CLEAR ALL** asks twice.



The same room from where you stand. The loudspeaker on the left is fed; the one beside it is switched off and is drawn solid, only unlit.

YOU CANNOT WALK THROUGH IT

ROOMS, AND YOUR BODY

A piece belongs to the room its centre is in, and stays inside it. Anything that blocks sound blocks you as well, so the sofa and the bookcase stop you walking; the rug and the curtain do not. If a piece is set down on top of you, you may walk out of it.

WHAT IT DOES TO THE SOUND

Four things, each from the acoustics tables or from the geometry, and each measured in the rendered audio.

ABSORPTION

PER OCTAVE BAND

Every piece brings its own equivalent absorption area, and it adds straight into the room's Eyring decay. Furnishing the tiled living room takes its reverberation time at 1 kHz from **3.42 s to 1.03 s**; the rendered decay measures 1.13 s. A rug replaces the floor it covers rather than adding to it.

BLOCKING

OVER, UNDER OR ROUND

A path that runs into a piece bends over, under or round it, with the doorways' edge-diffraction law. A bookcase between you and a source costs the direct sound **18.2 dB at 4 kHz**, matching a hand calculation; walking out from behind it the loss changes by at most **0.19 dB per centimetre**, so an edge never clicks.

THE READOUT LINE

Under the tray, for the room you are standing in: how much absorption its furniture adds at 1 kHz, and the reverberation time the engine now calculates. The room on the previous page, tiled and with its two doors open, reads RT60 **1.73 s** empty and **+9.1 m² · RT60 0.76 s** with those seven pieces in it.

SCATTERING

REFLECTIONS BROKEN UP

A furnished room returns its wall reflections less cleanly. What each bounce loses is not thrown away: it goes to the room's tail, the same route the STUDIO material uses.

A HARD TOP

TABLE, PIANO LID

The table's top and the closed piano lid are hard, and throw a reflection up at a head above them — the classic table bounce. It fades off the edge rather than vanishing. Only the table's top slab blocks, 71 to 75 cm up; the piano's body blocks from 35 cm to 1 m.

THE TEN PIECES

PIECE	W × D × H (M)	STANDS IN THE WAY
SOFA	2.10 × 0.90 × 0.85	yes
ARMCHAIR	0.85 × 0.85 × 0.90	yes
BED	2.05 × 1.60 × 0.55	yes, low
RUG	2.40 × 1.70 × 0.02	no — it replaces the floor
CURTAIN	2.40 × 0.15 × 2.50	no — it kills a wall
BOOKCASE	1.00 × 0.35 × 2.00	yes; also scatters
TABLE	1.60 × 0.90 × 0.75	its top only; hard
GRAND PIANO	2.10 × 1.50 × 1.00	its body; lid hard
WARDROBE	1.20 × 0.60 × 2.10	yes, tall
PERSON	0.50 × 0.30 × 1.75	yes; absorbs the top

Hover a piece in the tray for what it is and why it sounds the way it does. The layout is saved with the project and in patch files; it is not a set of host parameters, so it is not automated.

HONESTLY

Everything is geometric acoustics. A sofa in a corner does not damp a bass mode here, and nothing changes when a person sits down.

YOUR OWN PICTURES ON THE WALLS

Load an image, click a wall to hang it. As a print it is only a picture; as an acoustic panel it is heard.

HANGING ONE

LOAD, THEN CLICK A WALL

LOAD under **Wall pictures** picks a JPEG or PNG. The panel shrinks it to 1024 px, keeps it with the project, and waits for you to click a wall — in the plan or in the 3D view. Click a picture to select it; drag it along its wall, and in the 3D view up and down too. It is kept on its wall, under the ceiling and clear of any door. **DELETE** takes the selected one down. Up to **eight**; the counter beside the buttons says how many hang.

SIZE AND FRAME

S / M / L / XL, WIDTH, FRAME

Quick widths: S 40 cm, M 70 cm, L 1 m, XL 1.5 m; or **WIDTH** anywhere from 20 cm to 2 m. The height follows the image's own proportions. With nothing selected they set the size of the next one you hang. **FRAME** is thin black, oak or gilt moulding, or an unframed canvas whose picture wraps round its edges — purely visual.

PRINT OR ACOUSTIC PANEL

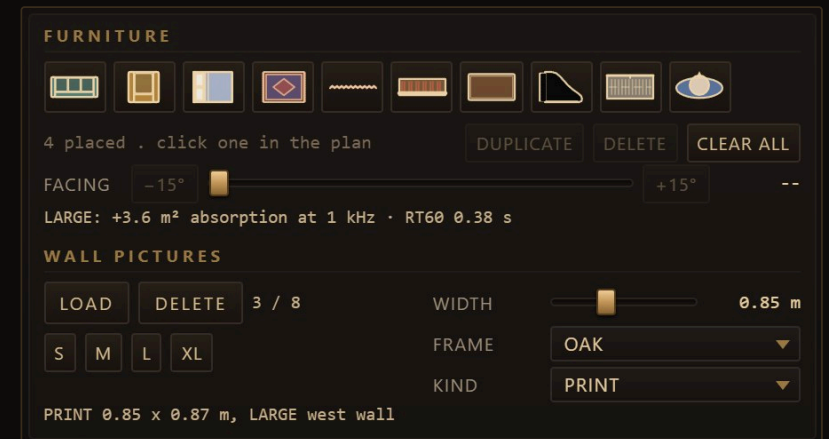
A **PRINT** is paper behind glass: acoustically it is part of the wall, and it changes nothing you hear. The panel says so.

An **ACOUSTIC PANEL** is the same picture printed on a 5 cm fabric absorber, the kind sold for studios. Its face area joins the room's absorption in place of the wall it covers: **4 m²** of them halve the tiled living room's decay at 1 kHz, **3.42 s to 1.69 s**.

Pictures are stored inside the project and in patch files, image and all, so a project opens complete on another computer. They are drawn in the plan, the 3D view, CINEMATIC and PHOTO, and in exported videos. On a broken wall a picture hangs on the panel under its centre, at that panel's angle.



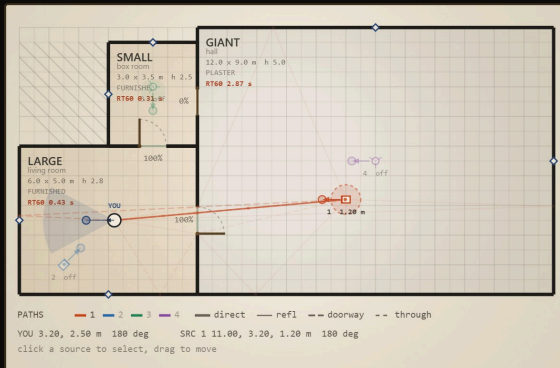
Three pictures on the west wall of the living room, above a stereo pair of loudspeakers.



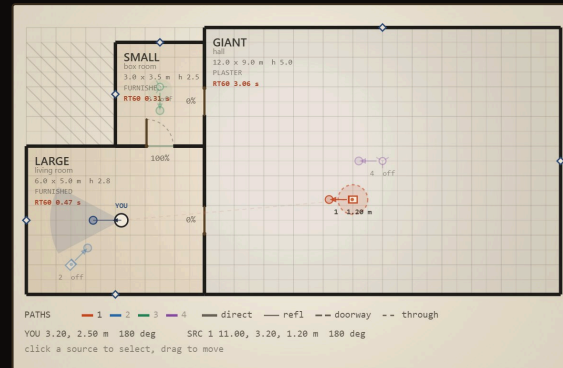
The Furniture group, with Wall pictures below the tray. The line at the foot names what is selected: here a print, 0.85 × 0.87 m, on the living room's west wall.

THREE DIFFERENT WAYS IN

A sound in the next room reaches you by several routes at once, and they do not sound alike. This is the part of the instrument the name is about.



Door open. The path bends at the edge of the opening and the sound is localised **at the doorway**, not at the source — exactly as it is in a real flat.



Door shut. What is left comes through the leaf and through the party wall, radiated from the point of each nearest straight line, so it still comes from the right direction.

ROUND THE FRAME

DIFFRACTION

Inside the open strip it is line of sight. Outside it, the path bends at the nearest point of the opening and pays Maekawa's attenuation, which costs high frequencies far more than low ones — the bent path loses **6 dB more at 4 kHz** than the clear one. The edge fades in rather than switching on, so walking past a doorway never clicks.

THROUGH THE LEAF

A LIGHT INTERIOR DOOR

A shut door is not a wall. It passes the mass law of a light leaf — **14 dB** of loss at 125 Hz rising to **32 dB** at 8 kHz. Shutting it drops 1 kHz by **20 dB** and darkens what is left: 4 kHz loses **8 dB more** than 250 Hz. That is the sound of a party next door.

THROUGH THE WALL

THE PARTY WALL

Heavier: **30 dB** at 125 Hz to **53 dB** at 8 kHz. It is usually below the door, but with every door shut it is what you are left with — measured **24 dB** below the same source at the same distance in your own room. Bass, and almost nothing else.

A door part open sits between the two, continuously: the engine does not crossfade a shut sound into an open one. **Walking through** an open door is continuous too: the doorway carries the room behind you to second order, and for 0.4 m either side of the plane both rooms are heard and handed over. Walking into the source's room holds about -24.5 dB through the plane, where it used to step 2 dB down.

WHY THERE IS NO REVERB AMOUNT

Each room runs its own sixteen-line reverberator, with delays from that room's mean free path and a decay per octave band from its materials, its open doors and the air in it. The rooms are coupled through the doors and walls with the energy balance of coupled-room theory, so the hall's long tail genuinely arrives in the living room.

Inside every one of those lines sits a chain of **allpass diffusers**. An allpass has unity gain at every frequency, so it cannot change the decay — but its length counts towards the loop, which decides whether the network's resonances are close enough together to hear as a room rather than as ringing, and it splits each echo into a train, which decides whether the tail has grain. A perfectly diffuse field has a spectral roughness of **5.57 dB**; this one measures **5.6 to 6.6**, where before the diffusers it measured **6.1 to 10.7**.

What that field is **worth**, relative to the direct sound, is not a taste decision. For a diffuse field it is fixed by the room's total absorption, and the engine calibrates it to exactly that — so the direct-to-reverberant ratio, your ears' strongest distance cue, comes out right by construction. Walk twice as far away and it falls by **6.2 to 7.0 dB**; walk in to half the distance and it rises by **3.6 to 5.5**. At the critical distance it reads between **−1.3 and −0.4 dB** against the **+0.9 dB** of the head's own frontal gain, for every one of the five materials.

That is why there is no “reverb amount”: turning one up would lie about how far away you stood. You get three trims instead — DIRECT, EARLY and REVERB — each at 0 dB, for when you want the lie on purpose.

THE DOUBLE SLOPE

A flat does not have one reverberation time, it has as many as it has rooms. Stand in the box room lined in drapes with the hall in tiles: its own field is **16 dB down inside 40 ms** and gone, then the hall's answer arrives through the doorway at the hall's own **5.8 seconds**. The decay falls off a cliff and then keeps going — what a small dead room inside a live building does, and a single reverb cannot. Shut the door and that slope goes with it.

```
YOU IN LARGE
> SRC 1 LARGE 1.57 m
  SRC 2 off
  SRC 3 off
  SRC 4 off
DRR      +0.0 dB
```

Top left of the 3D view: which room you are in, and for each of the four sources its room and its distance from you. An OFF source still gets a line, marked, because knowing which one is silent is half of what the readout is for.

```
IN -inf OUT -inf DRR 0.0
```

The right end of the foot: input and output level, and the direct-to-reverberant ratio in dB. Positive means you are inside the critical distance.

READING THE DRR

It counts every arrival that took the shortest route — including through a doorway or a shut leaf — as “direct”. That is the useful reading rather than the textbook one: it tells you how much of what you hear is the sound itself and how much is the building.

SEVEN CONTROLS AT THE END OF THE CHAIN

DIRECT

–24 TO +6 DB, DEFAULT 0

The straight line of sight. Pull it down and the room is all you have left — a way of moving a source further away without moving it.

REVERB

–24 TO +6 DB, DEFAULT 0

The coupled tails. The hall's arrives through whatever the doors are doing, so this and the doors do different jobs.

OUTPUT

–24 TO +6 DB, DEFAULT 0

The last thing in the chain. Two sources in one room and four in three can add up to more than one did, so there is somewhere to give it back.

HEAD

BUILT IN, OR A SOFA FILE

The ears you listen with. **Load SOFA** takes any measured head in the standard SOFA format (AES69) — your own, or one from a public set such as SADIE, ARI or HUTUBS — for live playing and every render; **Built in** goes back. It is kept with the project.

EARLY

–24 TO +6 DB, DEFAULT 0

The wall bounces and the doorway paths. This is the part that tells you the size and the shape of the room you are in.

MIX

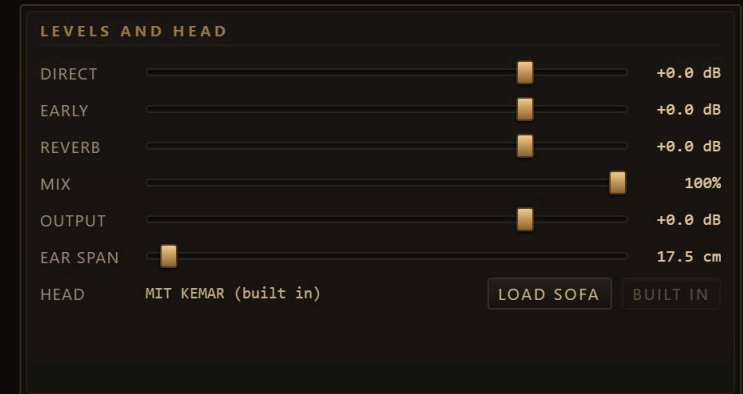
0 TO 100 %, DEFAULT 100

Dry against placed. At 100 you are entirely in the apartment. At 0 the input passes through untouched, exactly.

EAR SPACING

15 TO 100 CM, DEFAULT 17.5

How far apart your two ears sit. 17.5 cm is a head; a metre is a spaced pair of microphones, and the image opens out accordingly.



Levels and head. Rays, Default, Save and Load sit in the group beside it, View and files, with a readout of where you are standing (page 20).

WHAT EAR SPACING REALLY DOES

It stretches the interaural time difference and the two ear distances, leaving the pinna colouration alone — so it is a head on a wider neck, not a bigger head. Past about 25 cm it stops being a person and becomes a spaced microphone pair, which is a perfectly good thing to be: it widens the image without the phase trouble a stereo widener brings.

THE TEST SIGNAL AND THE STANDALONE

Walking around a room is not something you can do with one hand on a transport bar, so the plugin can play its own sound.

Load picks a WAV, AIFF, FLAC or MP3 off disk. **Play** loops it into the **MAIN** input — both channels — so every source fed from MAIN plays it, and you are free to spend both hands on the views. Sources fed from AUX are untouched, so you can audition one thing while a second track keeps running in another room.

Stop hands MAIN back to the host, immediately. The test signal is not a host parameter and is not saved in a patch: it is an audition tool, not part of the sound.

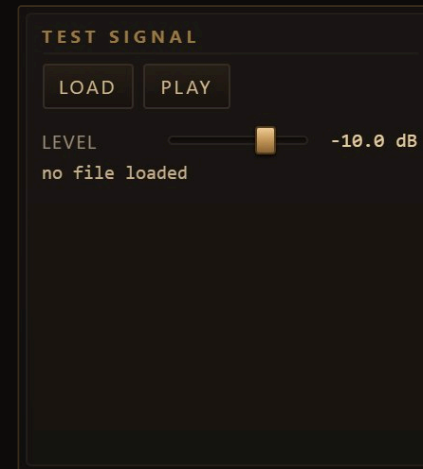
The same controls are in the VST3 and in the standalone. The standalone needs no host at all — open it, load a loop, and walk the flat.

PATCHES

Save and **Load** write and read a JSON file in `DOCUMENTS\BROKILD PATCHES\THIN WALLS`. A patch holds every surface, every broken wall, the doors, all four sources, the levels, the furniture and the pictures — the whole arrangement. **Default** puts every parameter back and empties the rooms; the floor plan itself never changes.

WHAT TO WALK AROUND TO

Something broadband and rhythmic. A shut door takes **20 dB** off 1 kHz and a good deal more off the top, so a loop with transients and some air in it tells you where you are far faster than a pad does — a drum loop, a spoken word, a strummed chord. Sustained, low-passed material is exactly the hardest case for the ear to place, and the flat will sound less impressive than it is.



The test signal group. The line underneath names the loaded file and its length.

AUTOMATION

All **65** parameters are host parameters, including every source position, every surface, every broken wall and your own position. Walking is recorded as automation, so a move you made by hand can be replayed exactly — and a source can be given a path by drawing one in your DAW.

A source moving at 2 m/s on a held tone keeps everything above 6 kHz **84 dB down**, and a wall moved under a sounding note **92 dB down**: the geometry changes continuously and nothing clicks.

CINEMATIC AND PHOTO

A second, much heavier renderer for the 3D view. It costs the graphics card and never the audio.



The ordinary view. Still draws nothing at all while nothing changes.



CINEMATIC, HIGH. Shadows from the lamp, ambient occlusion, bloom, film tone curve. The corner reads the frame time.



PHOTO, ULTRA, converged: light traced off the walls, soft shadows under the stands. 512 samples a pixel, 5.6 s.

CINEMATIC

THE HEADER BUTTON

An HDR picture with multisampling, real shadows from the pendants and the hall's window, ambient occlusion, physically based highlights, bloom and a filmic tone curve. It runs continuously while on — dust in the window light, a faint beam under each lamp. The first time it is switched on it compiles its shaders, a couple of seconds. Remembered on this computer, not in the project.

HIGH OR ULTRA

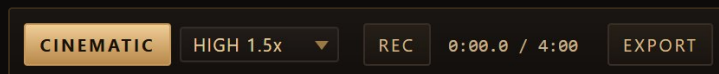
THE MENU BESIDE IT

HIGH renders at 1.5 times the screen's pixels, **ULTRA** at 2 times, both with 4× multisampling. About **2 ms** of graphics card per frame at HIGH on a current card; the corner of the view shows what yours takes. ULTRA also takes PHOTO further: 512 samples a pixel instead of 256.

PHOTO

WHEN YOU STAND STILL

Stand still for 0.6 s and PHOTO takes over: the light is traced, bouncing off the walls and through the open doors, and the picture refines itself over a few seconds. The corner counts it up to done — and then the graphics card rests. Move, and the live picture is back at once.



The header, CINEMATIC on: its quality, REC with the take clock, and EXPORT.

LAMPS THAT FOLLOW THE SOUND

Each room's pendants can follow the sound in that room, or pulse on your host's beat. At zero they are exactly the lamps they always were.

The engine measures how much sound is in each room: what that room's sources play plus its own late field, half by level and half by **punch** — a fast envelope over a slow one, so an onset flashes and a held sound glows. Measured, an onset reads **0.93** where the same noise held reads **0.44**; silence is exactly zero, and a sealed room next door stays at **0.000**. The lamps you see are the rooms you would hear.

SYNC

0 TO 100 %, DEFAULT 20

How much the lamps follow. At 0 they are off the hook entirely. The default breathes a few percent; all the way up the lamps dim in silence and flare on every hit.

FOLLOW

THE DEFAULT DRIVE

Each room's pendants follow the sound in that room. The picture is fed by the 30 Hz scene stream, so a hard transient can look a frame behind.

BEAT

THE HOST'S CLOCK

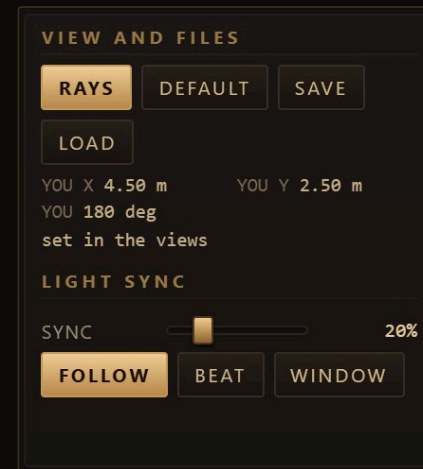
The pendants pulse on the host's beats while its transport plays. Stopped, or with no host clock, they follow the sound instead.

WINDOW

OFF BY DEFAULT

Lets the hall's daylight window join in. Daylight does not usually dance, which is why it waits to be asked.

Like CINEMATIC, light sync is remembered on this computer rather than in the project. It costs nothing on the audio thread; with the ordinary view it means the view redraws while sound plays instead of resting. An exported video uses the offline render's own light and the recorded host clock for every frame, so it pulses exactly with its sound.



View and files, with **Light sync** at its foot.

THE REST OF THE GROUP

Rays draws the engine's paths on the plan; **Default**, **Save** and **Load** are the patch buttons of page 18. The readout is where you stand and which way you face — set in the views, never here.

FILM A WALK THROUGH THE FLAT

Record a take of up to four minutes, then render it as an MP4 with its own sound.

REC

IN THE HEADER

Captures a take: the sound going in (the test signal, if it is playing), every host parameter as it moves, the furniture, and where you look. Walk the rooms, open doors, move sources — all of it is recorded. The button reads **STOP** while it runs and the clock beside it counts against the **4:00** limit. Press it again to stop; the take is held until you record another.

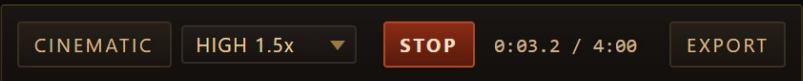
EXPORT

THE PANEL UNDER THE HEADER BUTTON

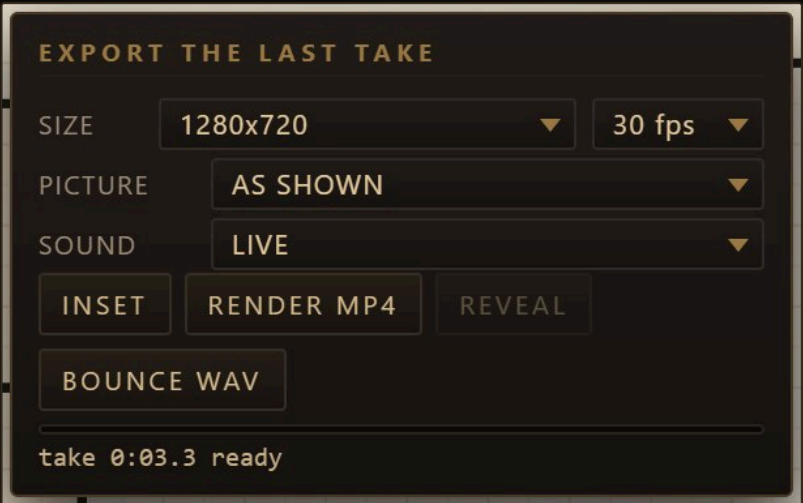
SIZE	1280×720, 1920×1080, 2560×1440 or 3840×2160
fps	30 or 60; 60 doubles the rendering time
PICTURE	AS SHOWN, CINEMATIC, PHOTO 16 or 64 — PHOTO traces that many samples into every frame
SOUND	LIVE, HIGH, ULTRA or ULTRA + BASS — how the take's sound is rendered (next page)
INSET	a small plan in the corner; off by default
RENDER MP4	renders the last take; CANCEL deletes the partial file
REVEAL	shows the last video in Explorer
BOUNCE WAV	the take's sound alone, at the SOUND chosen, as a 24-bit WAV beside the videos

WHY PICTURE AND SOUND CANNOT DRIFT

The sound is re-rendered offline through a fresh engine, and every frame is redrawn from the same recording, so picture and sound share one timeline. H.264 with AAC sound, in DOCUMENTS\THIN WALLS VIDEOS.



Recording. REC has become STOP, and the take is 3.2 seconds old.



The export panel, with a take held and SOUND at ULTRA + BASS. Until there is one, the line at its foot says what is missing; afterwards it names the last video written.

WHAT IT IS FOR

Showing someone what a room does — a walk out of the living room and into the hall, the door closing behind you — with the sound it actually makes. The panel is locked while an export runs.

FOUR WAYS TO HEAR A TAKE

Offline there is no audio thread to keep up with. None of this changes what you hear while you play, or what it costs.

LIVE

WHAT YOU HEARD

The live engine, sample for sample: reflections to the second order, a statistical late field per room.

ULTRA

6TH ORDER, AND TRACED TAILS

The statistical reverb is replaced by one traced through the actual rooms: rays bounce off the folded walls, the furniture, the rug and the doors as they stand, and what crosses a sphere round your head becomes the tail, band by band and direction by direction. Each room rings with its own shape, and the hall next door comes round the door — or through the wall.

HIGH

REFLECTIONS TO THE 4TH ORDER

Five times as many wall reflections (128 against 24 in the tiled living room) and twice the distinct early echoes; the late field hands over exactly what they now carry.

ULTRA + BASS

BELOW 220 HZ, A REAL WAVE

Under the crossover the whole apartment is simulated as air on a 10 cm grid: room modes, bass bending round a door, a door opening mid-take. Sound through walls and leaves still comes from the measured transmission losses.

EXPORT THE LAST TAKE

SIZE **1280x720** **30 fps**

PICTURE **AS SHOWN**

SOUND **LIVE**

INSET **RENDER MP4** **REVEAL**

BOUNCE WAV

take 0:03.3 ready

SOUND, for the MP4 and BOUNCE WAV alike.

WHAT IT COSTS

Only time, and only when you render: on a many-core desktop a four-second take takes about 6 s at HIGH or ULTRA and 10 s at ULTRA + BASS. A take that moves is traced again every 25 cm.

HONESTLY

Rays know nothing of room modes — that is what ULTRA + BASS is for. Below its crossover the wave is DIRECT, EARLY and REVERB at once, so those three do not reach there.

CHECKED AGAINST THE LIVE ENGINE

The traced tail lands within **0.4 to 0.9 dB** of the statistical one in four materials, decays within Eyring's times, and puts the hall's tail **60 dB** louder into a dead room once the door opens. Through a shut wall it is within **0.6 dB**. The wave gives the living room's axial modes at **28.6** and **34.3 Hz**, where $c / 2L$ puts them, and its direct sound lands within **0.2 dB** of the engine's. Above the crossover ULTRA + BASS changes 1 kHz by **0.02 dB**.

WHAT WAS MEASURED

The bench renders real audio and measures it — 90 checks, 26 for furniture, pictures and light and 33 for the offline renders, all clear, beside 141 that drive the panel. These are its numbers, not estimates.

ITEM	MEASURED
Interaural delay at 90°	−708 μs (Woodworth's formula gives 666)
Front against back	5.0 dB apart in spectral shape at 4 and 8 kHz, 1.0 dB in level — the pinna
Level with distance	6.1 dB per doubling — and the interaural level difference grows 4.6 dB by 0.3 m, out of the geometry
Direct path	within a tenth of a decibel of the physics from 10 Hz to 16 kHz, wherever a source stands
Air absorption	8 kHz costs 0.73 dB more than 250 Hz over 11 m (ISO 9613-1: 0.78)
Reverberation time	against Eyring in all fifteen room and material cases; the tail's spectral roughness 5.6 to 6.6 dB, against the 5.57 dB of a perfectly diffuse field
Direct-to-reverberant	−1.3 to −0.4 dB at the critical distance against the head's own +0.9; +3.6 / −7.0 at half and double
Shutting a door	−23 dB at 1 kHz, and 4 kHz loses 12 dB more than 250 Hz
Party wall and shut door	31 dB below the same distance in your own room
Moving a source	2 m/s on a pure tone: everything above 6 kHz stays 84 dB down
MIX at 0	bit-exact passthrough
Furnishing	tiled living room, 1 kHz: 3.42 s to 1.03 s by Eyring, 1.13 s in the rendered decay
A bookcase in the way	18.2 dB off the direct sound at 4 kHz; at most 0.19 dB per cm at its edge

ITEM	
Format	Windows VST3 (64-bit), plus a standalone application
Type	Effect — stereo in and out, plus “Aux In”, the host's sidechain
Parameters	65, all automatable
Materials	Five, per surface: nine choices. Two walls of every room break and push ±0.6 m
Head	MIT KEMAR compact set, 368 directions, diffuse-field equalised, minimum phase with a separate onset delay; or any SOFA file
Reflections	image sources to second order (4th and 6th offline); a general surface-sequence search once a wall is broken
Reverberator	one 16-line feedback delay network per room, coupled through the doors and walls. Offline: tails traced through the rooms, and a wave simulation below 220 Hz
Delay reader	16-tap windowed sinc
Furniture	ten pieces, up to 24; up to eight pictures, print or acoustic panel
Picture	ordinary, CINEMATIC, PHOTO; light sync
Video	takes up to 4 minutes; MP4 (H.264 + AAC), 720p to 4K, 30 or 60 fps
CPU	one source about 18 % of a core at 48 kHz; four sources and ten pieces of furniture, about 44 %. The picture runs on the graphics card
Sample rates	any; bench-tested at 44.1 and 96 kHz. Hosts: any VST3 host, developed against Ableton Live 12
Price	free. No account, no registration, no telemetry.

The head data is the MIT KEMAR HRTF measurements of [W. G. Gardner and K. D. Martin \(1994\)](#), MIT Media Lab, used with thanks. Absorption coefficients from the standard tables (Kuttruff; Vorländer). Diffraction after Maekawa; air absorption after ISO 9613-1.

HONEST LIMITS

WHAT IT IS NOT

The floor plan is fixed. One apartment, three rooms, three doors. Two walls of each room can be broken and pushed, so the rooms are not stuck as boxes — but where the rooms are, and which doors join them, you cannot change. A second plan would be a different instrument.

Reflections are specular and go to second order. Only STUDIO scatters, and there is no diffraction round the edges of the reflections themselves. Beyond the second bounce the room's own reverberator takes over, which is where that energy belongs anyway.

Two sources in one room share its field. That is correct — a room has one reverberant field — but the share each hands it is taken at 1 kHz rather than per band.

Furniture is geometric. It absorbs, scatters, blocks and reflects; it does not damp a room's bass modes, and a print on the wall does nothing you can hear. PHOTO traces a simplified box model of the flat, so what it lights is the rooms and their furniture, not every detail of the ordinary view.

Breaking a wall moves the early reflections, not the modes. The late field here is statistical, so splaying a wall does not redistribute a room's standing waves; above the Schroeder frequency that is where splay works in a real room too.

The DRR readout is the useful reading, not the textbook one. It counts an arrival that came through a doorway or a shut leaf as direct, because that is the sound rather than the building. And the test signal **replaces** the MAIN input while it plays, rather than mixing with it; AUX carries on regardless.

None of these is a placeholder. They are the edges of what was built, written down so you know where they are.

BROKILD

Free, and free to pass on.
peterboggild.github.io/BrokildApps

BUILD 260925.5 · WINDOWS VST3

MIND THE NEIGHBOURS

